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Liquid feeding unit, and liquid chromatography analysis system and control method thereof

Abstract

A first solution is supplied from a first pump. A second solution is supplied from a second pump. A flow path from the first pump and the second pump to a column is switched between a first flow path and a second flow path. In the first flow path, a first mixer is located upstream of an injection part for a sample, and the second mixer is located downstream of the injection part. In the second flow path, the first mixer and the second mixer are located upstream of the injection part. The first flow path is formed in a first mode in which the sample is diluted before introduction into the column. The second flow path is formed in a second mode in which the sample is not diluted before introduction into the column.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to a liquid chromatography analysis system that provides a mobile phase according to a gradient elution method.

Description of the Background Art

(2) The gradient elution method is an analysis method of using a compound liquid of a plurality of solvents as a mobile phase in a liquid chromatography analysis. Some conventional systems are configured to supply a mobile phase according to the gradient elution method. Some conventional liquid chromatography analysis systems are configured to dilute a sample before introduction into a column.

(3) Techniques regarding the above systems are disclosed in, for example, PTL 1 (U.S. Patent

Application Publication No. 2002/117447), PTL 2 (U.S. Patent Application Publication No. 2004/035789), PTL 3 (U.S. Patent Application Publication No. 2008/264848), and PTL 4 (U.S. Patent Application Publication No. 2010/176043).

(4) However, if the conventional systems are provided with a configuration for diluting a sample before introduction into a column, they are not suitable for an analysis in a mode in which the sample is not diluted before introduction into the column.

(5) Thus, such a technique is required that adapts a liquid chromatography analysis system to both of an analysis in a mode in which the sample is diluted before introduction into the column and an analysis in a mode in which the sample is not diluted before introduction into the column.

SUMMARY OF THE INVENTION

(6) A liquid feeding unit according to an aspect of the present disclosure is a liquid feeding unit that feeds a first solution and a second solution as a mobile phase to a column of a liquid chromatography analysis system. The first solution is supplied from a first pump. The second solution is supplied from a second pump. The liquid chromatography analysis system includes an injection part into which a sample is injected. The liquid feeding unit includes a first mixer, a second mixer, and a switch device that switches a flow path from the first pump and the second pump to the column between a first flow path and a second flow path. The first mixer and the second mixer are included in each of the first flow path and the second flow path. In the first flow path, the first mixer is located upstream of the injection part, and the second mixer is located downstream of the injection part. In the second flow path, the first mixer and the second mixer are located upstream of the injection part. The switch device forms the first flow path as the flow path in a first mode in which the sample is diluted before introduction into the column. The switch device forms the second flow path as the flow path in a second mode in which the sample is not diluted before introduction into the column.

(7) A liquid chromatography analysis system according to an aspect of the present disclosure is a liquid chromatography analysis system that uses a first solution and a second solution as a mobile phase. The liquid chromatography analysis system includes a column, a first pump that supplies the first solution, a second pump that supplies the second solution, a sampler that injects a sample toward the column, a first mixer, a second mixer, and a switch device that switches a flow path from the first pump and the second pump to the column between a first flow path and a second flow path. The first mixer and the second mixer are included in each of the first flow path and the second flow path. In the first flow path, the first mixer is located upstream of an injection part into which the sample is injected by the sampler, and the second mixer is located downstream of the injection part. In the second flow path, the first mixer and the second mixer are located upstream of the injection part. The switch device forms the first flow path as the flow path in a first mode in which the sample is diluted before introduction into the column. The switch device forms the second flow path as the flow path in a second mode in which the sample is not diluted before introduction into the column.

(8) A control method for a liquid chromatography analysis system according to an aspect of the present disclosure is a control method for a liquid chromatography analysis system that uses a first solution and a second solution as a mobile phase, and the control method is implemented by a computer. The control method includes: obtaining an analysis mode; and controlling a flow path from a first pump and a second pump to a column of the liquid chromatography analysis system in accordance with the analysis mode, the first pump supplying the first solution, the second pump supplying the second solution. The liquid chromatography analysis system includes an injection part into which a sample is injected, a first mixer, and a second mixer. The controlling the flow path includes, when the analysis mode is a first mode in which the sample is diluted before introduction into the column, controlling the flow path such that the first mixer is located upstream of the injection part and the second mixer is located downstream of the injection part. The controlling the flow path includes, when the analysis mode is a second mode in which the sample is not diluted

before introduction into the column, controlling the flow path such that the first mixer and the second mixer are located upstream of the injection part.

(9) The foregoing and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a schematic configuration of an LC system **100** of the present disclosure.
- (2) FIG. 2 shows a position of a liquid feeding unit **13** in a standard mode.
- (3) FIG. 3 shows a position of liquid feeding unit **13** in a dilution mode.
- (4) FIG. 4 shows a specific configuration example of a liquid feeding unit according to an existing technique.
- (5) FIG. 5 shows a specific configuration example of a liquid feeding unit according to an existing technique.
- (6) FIG. 6 shows a specific configuration example of a liquid feeding unit according to an existing technique.
- (7) FIG. 7 shows a specific configuration example of a liquid feeding unit according to an existing technique.
- (8) FIG. 8 shows evaluations of a configuration of an embodiment and each of configurations of the existing techniques in terms of five criteria.
- (9) FIG. 9 shows a specific configuration example of liquid feeding unit **13**.
- (10) FIG. 10 shows a connection pattern example of ports **601** to **606** with a flow path switching valve **600** in a standard position.
- (11) FIG. 11 shows a connection pattern example of ports **601** to **606** with flow path switching valve **600** in a dilution position.
- (12) FIG. 12 is a flowchart of a process performed for analyzing a sample by LC system **100**.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(13) A liquid chromatography analysis system (referred to as “LC system” below) according to an embodiment of the present disclosure will be described below in detail with reference to the drawings. The same or corresponding parts in the drawings have the same reference characters allotted, and description thereof will not be repeated.

(14) [Schematic Configuration of Liquid Chromatography Analysis System]

(15) FIG. 1 shows a schematic configuration of an LC system **100** of the present disclosure. As shown in FIG. 1, LC system **100** includes a liquid feeder **10**, an autosampler **20**, a column oven **30**, a detector **40**, an analysis flow path **50**, and a controller **60**.

(16) LC system **100** is configured to supply a mobile phase according to a gradient elution method. In LC system **100**, a first solvent and a second solvent are prepared as solvents constituting the mobile phase. Liquid feeder **10** is connected with a tank **71** that contains the first solvent and a tank **72** that contains the second solvent. The first solvent and the second solvent are different from each other in dilution power. In one implementation, the first solvent is water, and the second solvent is methanol. Liquid feeder **10** mixes the first solvent and the second solvent to supply analysis flow path **50** with the mobile phase according to the gradient elution method.

(17) Liquid feeder **10** includes a first pump **11** that supplies the first solvent to analysis flow path **50** and a second pump **12** that supplies the second solvent to analysis flow path **50**. Controller **60** controls the respective flow velocities of first pump **11** and second pump **12** to adjust the respective flow rates of the first solvent and the second solvent in the mobile phase, thereby adjusting a ratio between the first solvent and the second solvent in the mobile phase.

(18) Liquid feeder **10** further includes a liquid feeding unit **13** that mixes the first solvent supplied from first pump **11** and the second solvent supplied from second pump **12**. The configuration of liquid feeding unit **13** will be described below in detail with reference to FIGS. **2** and **3**.

(19) Autosampler **20** injects a sample into analysis flow path **50**.

(20) Column oven **30** includes a column **31**. Column **31** is used to separate the sample injected into analysis flow path **50** by autosampler **20**. Column **31** is accommodated in column oven **30**. Column oven **30** controls the temperature of column **31** to a preset temperature.

(21) Detector **40** is a device for detecting one or more components separated from the sample in column **31**. Detector **40** obtains a detection signal based on each of the one or more components separated in column **31** and transmits the detection signal to controller **60**.

(22) Controller **60** controls operations of liquid feeder **10**, autosampler **20**, and column oven **30** and creates various mathematical operations and chromatograms based on the detection signal obtained by detector **40**.

(23) Controller **60** includes a processor **61**, a storage device **62**, and an interface **63**. Processor **61** performs the control described above and mathematical operations for creating the chromatograms. Storage device **62** stores a program and data for the above mathematical operations. Interface **63** functions as an interface for communications between processor **61** and each of liquid feeder **10**, autosampler **20**, and column oven **30**.

(24) [Standard Mode and Dilution Mode]

(25) A sample may be diluted in order to adjust a peak shape of a chromatogram created for the sample. Herein, in LC system **100**, an analysis mode when a sample is analyzed without dilution is referred to as “standard mode”, and an analysis mode when a diluted sample is analyzed is referred to as “dilution mode”.

(26) “Dilution mode” is an example of the first mode in which the sample is diluted before introduction into column **31**. “Standard mode” is an example of the second mode in which the sample is not diluted before introduction into column **31**.

(27) [Schematic Configuration of Liquid Feeding Unit]

(28) In LC system **100**, liquid feeding unit **13** takes positions for forming different paths between the standard mode and the dilution mode. FIG. **2** shows a position of liquid feeding unit **13** in the standard mode. The position shown in FIG. **2** is also referred to as “standard position”. FIG. **3** shows a position of liquid feeding unit **13** in the dilution mode. The position shown in FIG. **3** is also referred to as “dilution position”.

(29) As shown in FIGS. **2** and **3**, liquid feeding unit **13** includes a first mixer **13G** and a second mixer **13D**. Each of first mixer **13G** and second mixer **13D** includes a mixing chamber that contains liquids supplied from a plurality of supply sources. The mixing chamber has an opening for discharging the liquids supplied from the plurality of supply sources in a mixed state. In one implementation, each of first mixer **13G** and second mixer **13D** is implemented by a gradient mixer manufactured by Shimadzu Corporation (https://www.an.shimadzu.co.jp/hplc/prominence/modules/8_option.htm). The mixing capacity of each of first mixer **13G** and second mixer **13D** is based on the capacity of its mixing chamber.

(30) In the example of FIG. **2**, liquid feeding unit **13** includes pipes **131** to **134**. In the example of FIG. **2**, pipes **131** to **134** constitute a flow path, which extends from first pump **11** and second pump **12** to column **31**, of analysis flow path **50**.

(31) In the example of FIG. **3**, liquid feeding unit **13** includes pipes **131**, **132**, and **134** to **136**. In the example of FIG. **3**, pipes **131**, **132**, and **134** to **136** constitute a flow path, which extends from first pump **11** and second pump **12** to column **31**, of analysis flow path **50**.

(32) (FIG. **2**: Standard Position)

(33) At the standard position shown in FIG. **2**, the first solvent supplied from first pump **11** is introduced through pipe **131** into second mixer **13D**, and then introduced through pipe **133** into first mixer **13G**. On the other hand, the second solvent supplied from second pump **12** is introduced

through pipe **132** into first mixer **13G**. The first solvent and the second solvent are mixed in first mixer **13G**, and then introduced through pipe **134** and pipe **135** into column **31**.

(34) In FIG. **2**, an injection part for the sample from autosampler **20** in analysis flow path **50** is indicated as “autosampler **20**”. In FIG. **2**, the sample injected from autosampler **20** is introduced through pipe **135** into column **31**.

(35) At the standard position shown in FIG. **2**, the first solvent supplied from first pump **11** flows through second mixer **13D** and then first mixer **13G**, and then through (the injection part of) autosampler **20**. In other words, first mixer **13G** and second mixer **13D** are located upstream of (the injection part of) autosampler **20**.

(36) FIG. **2** shows a supplement N1 for illustrating a liquid feeding pattern in the standard mode. Supplement N1 includes a graph indicating a pattern of liquid feeding to analysis flow path **50** before and after injection of the sample from autosampler **20**. In the graph, the vertical axis schematically represents a flow rate of each solution at a sample introduction port of analysis flow path **50**, and the horizontal axis represents a time.

(37) As indicated in the graph of supplement N1, at a time **T0**, driving of first pump **11** and second pump **12** is started. Thus, supply of the first solvent and the second solvent to analysis flow path **50** is started. Subsequently, at a time **T1** to a time **T2**, the sample is introduced from autosampler **20** to analysis flow path **50**. Specifically, a valve located in autosampler **20** is switched, and the sample filling a sample loop also located in autosampler **20** is introduced into analysis flow path **50** at a flow rate set at time **T0**.

(38) When the introduction of the sample is complete at time **T2**, though the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter is constant in FIG. **2**, in actuality, the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter may change based on settings (settings according to the gradient elution method) adopted in the analysis of the sample.

(39) (FIG. **3**: Dilution Position)

(40) At the dilution position shown in FIG. **3**, the first solvent supplied from first pump **11** is introduced through pipe **131** into second mixer **13D**. On the other hand, the second solvent supplied from second pump **12** is introduced through pipe **132** into second mixer **13G**, and then introduced through pipe **134** and pipe **135** into second mixer **13D**.

(41) At the dilution position shown in FIG. **3**, the first solvent and the second solvent are mixed in second mixer **13D**, and then introduced through pipe **136** into column **31**.

(42) In FIG. **3**, an injection part for the sample from the autosampler **20** in analysis flow path **50** is indicated as “autosampler **20**” as in FIG. **2**. At the dilution position shown in FIG. **3**, the sample injected from autosampler **20** is introduced through pipe **135** into second mixer **13D**. Subsequently, the sample is diluted with the first solvent in second mixer **13D**, and then introduced through pipe **136** into column **31**.

(43) At the dilution position shown in FIG. **3**, the second solvent supplied from second pump **12** flows through first mixer **13G** and then (the injection part of) autosampler **20**, and then through second mixer **13D**. In other words, first mixer **13G** is located upstream of (the injection part of) autosampler **20**, and second mixer **13D** is located downstream of (the injection part of) autosampler **20**.

(44) FIG. **3** shows a supplement N2 for illustrating a liquid feeding pattern in the dilution mode. Supplement N2 includes a graph indicating a pattern of liquid feeding to analysis flow path **50** before and after injection of the sample from autosampler **20**. In the graph, the vertical axis schematically represents a flow rate, and the horizontal axis represents a time.

(45) As indicated in the graph of supplement N2, at time **T0**, driving of first pump **11** and second pump **12** is started. Thus, supply of the first solvent and the second solvent to analysis flow path **50** is started. Subsequently, at time **T1** to time **T2**, a valve located in autosampler **20** is switched, and the sample filling the sample loop also located in autosampler **20** is pushed out by the second

solvent fed through driving of second pump **12**, so that the sample is introduced into analysis flow path **50**. Thus, the pushed-out sample is diluted with the first solvent fed through driving of first pump **11**, and the diluted sample is introduced into column **31**.

(46) When the introduction of the sample is complete at time **T2**, a compound liquid of the first solvent and the second solvent mixed at a prescribed ratio is supplied to flow path **50**. Though the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter is constant in FIG. **3**, in actuality, the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter may change based on settings (settings according to the gradient elution method) adopted in analysis of the sample.

Specific Configuration Examples According to Existing Techniques

(47) Each of FIGS. **4** to **7** shows a specific configuration example of a liquid feeding unit according to an existing technique. Each specific example will be described below.

(48) (FIG. **4**: Bypass Scheme)

(49) The configuration example of FIG. **4** includes a first mixer **913G** and a second mixer **913D**. In FIG. **4**, a first pump **911**, a second pump **912**, pipes **811**, **812**, **813**, **814**, **815**, **816**, an autosampler **920**, and a column oven **930** are further included. Column oven **930** includes a column **931**.

(50) In the configuration example of FIG. **4**, the first solvent supplied from first pump **911** is introduced through pipe **811** into first mixer **913G**. The second solvent supplied from second pump **912** is introduced through pipe **812** into first mixer **913G**. The first solvent and the second solvent are mixed in first mixer **913G**, and then introduced through pipe **813** and pipe **816** into column **931**.

(51) The sample injected from autosampler **920** is introduced through pipe **815** into second mixer **913D**.

(52) In the configuration example of FIG. **4**, pipe **814** functions as a bypass line. In other words, a part of the compound liquid (the first solvent and the second solvent) in pipe **813** is introduced through pipe **815** into second mixer **913D**, and the rest is introduced through pipe **814** into second mixer **913D**.

(53) The compound liquid introduced through pipe **814** into second mixer **913D** dilutes the sample from autosampler **920**. The diluted sample is introduced from second mixer **913D** through pipe **816** into column **931**,

(54) FIG. **4** shows a supplement N3 for illustrating a liquid feeding pattern in the dilution mode. Supplement N3 includes a graph indicating a pattern of liquid feeding to the analysis flow path (pipes **811** to **816**) before and after injection of the sample from autosampler **920**. In the graph, the vertical axis schematically represents a flow rate, and the horizontal axis represents a time.

(55) As indicated in the graph of supplement N3, at time **T0**, driving of first pump **911** and second pump **912** is started. Thus, supply of the first solvent and the second solvent to the analysis flow path is started. Subsequently, at time **T1** to time **12**, the sample filling a sample loop connected with a valve switched in autosampler **920** is introduced into the analysis flow path. The sample is introduced into column **931** while being diluted with the compound liquid or the first solvent and the second solvent introduced through pipe **814** into second mixer **913D**.

(56) When the introduction of the sample is complete at time **T2**, the compound liquid of the first solvent and the second solvent is supplied to the analysis flow path as the mobile phase. Though a flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter is constant in FIG. **4**, in actuality, the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter may change based on settings (settings according to the gradient elution method) adopted in analysis of the sample.

(57) (FIG. **5**: Diluting Solution Introduction Scheme)

(58) The configuration example of FIG. **5** includes first mixer **913G** and second mixer **913D**. In FIG. **5**, first pump **911**, second pump **912**, a third pump **990**, pipes **821**, **822**, **823**, **824**, **825**, **826**, autosampler **920**, and column oven **930** are further included. Column oven **930** includes column **931**. Third pump **990** supplies a solvent (diluting solvent) for diluting the sample to the analysis

flow path.

(59) In the configuration example of FIG. 5, the first solvent supplied from first pump **911** is introduced through pipe **821** into first mixer **913G**. The second solvent supplied from second pump **912** is introduced through pipe **822** into first mixer **913G**. The first solvent and the second solvent are mixed in first mixer **913G**, and then introduced through pipe **823**, pipe **824**, and pipe **826** into column **931**.

(60) The sample introduced from autosampler **920** is introduced through pipe **824** into second mixer **913D**. Third pump **990** supplies the diluting solvent through pipe **825** to second mixer **913D**. The sample is diluted in second mixer **913**, and then introduced through pipe **826** into column **931**.

(61) FIG. 5 shows a supplement N4 for illustrating a liquid feeding pattern in the dilution mode. Supplement N4 includes a graph indicating a pattern of liquid feeding to the analysis flow path (pipes **821** to **826**) before and after injection of the sample from autosampler **920**. In the graph, the vertical axis schematically represents a flow rate, and the horizontal axis represents a time.

(62) As indicated in the graph of supplement N4, at time T0, driving of first pump **911** and second pump **912** is started. Thus, supply of the first solvent and the second solvent to the analysis flow path is started.

(63) Then, at time T1 to time T2, the sample is introduced from autosampler **920** to second mixer **913D** with reduced flow velocities of first pump **911** and second pump **912**. At time T1 to time T2, third pump **990** is driven, so that the diluting solvent is introduced into second mixer **913D**. The sample is diluted in second mixer **913D**, and then introduced through pipe **826** into column **931**.

(64) When the introduction of the sample is complete at time T2, driving of third pump **990** is stopped, and the flow rates of first pump **911** and second pump **912** are returned to the flow rates before the reduction. The compound liquid of the first solvent and the second solvent is thus supplied to the analysis flow path as the mobile phase. Though the flow rate ratio between the first solvent and the second solvent at time T2 and thereafter is constant in FIG. 5, in actuality, the flow rate ratio between the first solvent and the second solvent at time T2 and thereafter may change based on settings (settings according to the gradient elution method) adopted in analysis of the sample.

(65) (FIG. 6: Sample Solution Introduction Scheme)

(66) The configuration example of FIG. 6 includes first mixer **913G** and second mixer **913D**. In FIG. 6, first pump **911**, second pump **912**, third pump **990**, pipes **831**, **832**, **833**, **834**, **835**, **836**, autosampler **920**, and column oven **930** are further included. Column oven **930** includes column **931**. Third pump **990** supplies the solvent (diluting solvent) for diluting the sample to the analysis flow path.

(67) In the configuration example of FIG. 6, the first solvent supplied from first pump **911** is introduced through pipe **831** into first mixer **913G**. The second solvent supplied from second pump **912** is introduced through pipe **832** into first mixer **913G**. The first solvent and the second solvent are mixed in first mixer **913G**, and then introduced through pipe **833** and pipe **836** into column **931**.

(68) The sample injected from autosampler **920** is introduced through second mixer **913D** and pipe **836** into column **931** while being diluted with the diluting solvent supplied from third pump **990** through pipe **834**.

(69) FIG. 6 shows a supplement N5 for illustrating a liquid feeding pattern in the dilution mode. Supplement N5 includes a graph indicating a pattern of liquid feeding to the analysis flow path (pipes **831** to **836**) before and after injection of the sample from autosampler **920**. In the graph, the vertical axis schematically represents a flow rate, and the horizontal axis represents a time.

(70) As indicated in the graph of supplement N5, at time T0, driving of first pump **911** and second pump **912** is started. Thus, supply of the first solvent and the second solvent to the analysis flow path is started.

(71) Subsequently, at time T1 to time T2, pump **990** is driven with reduced flow rates of first pump **911** and second pump **912**, so that the sample is introduced from autosampler **920** to second mixer

913D. Thus, the sample is diluted with the first solvent from first pump **911** and the second solvent from second pump **912**, and then introduced into column **931**. At time **T1** to time **T2**, the ratio between the first solvent and the second solvent may be similar to a ratio at time **T0** or may be set to a different ratio. In other words, it suffices that this ratio is a ratio at which the sample is appropriately introduced from autosampler **920** into second mixer **913D**, and is not particularly limited.

(72) When the introduction of the sample is complete at time **T2**, driving of third pump **990** is stopped, and driving of first pump **911** and second pump **912** is restarted. Thus, the compound liquid of the first solvent and the second solvent is supplied to the analysis flow path as the mobile phase. Though the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter is constant in FIG. **6**, in actuality, the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter may change based on settings (settings according to the gradient elution method) adopted in analysis of the sample.

(73) (FIG. **7**: High-Pressure Gradient Dilution Scheme)

(74) The configuration example of FIG. **7** includes a mixer **913X**. In FIG. **7**, first pump **911**, second pump **912**, pipes **841**, **842**, **843**, **844**, autosampler **920**, and column oven **930** are further included. Column oven **930** includes column **931**.

(75) In the configuration example of FIG. **7**, the first solvent supplied from first pump **911** is introduced through pipe **841** into mixer **913X**. The second solvent supplied from second pump **912** is introduced through pipe **842** and pipe **843** into mixer **913X**. The first solvent and the second solvent are mixed in mixer **913X**, and then introduced through pipe **844** into column **931**.

(76) The sample introduced from autosampler **920** is introduced through pipe **843** into mixer **913X**. The sample is diluted with the first solvent in mixer **913X**, and then introduced through pipe **844** into column **931**.

(77) FIG. **7** shows a supplement **N6** for illustrating a liquid feeding pattern in the dilution mode. Supplement **N6** includes a graph indicating a pattern of liquid feeding to the analysis flow path (pipes **841** to **844**) before and after injection of the sample from autosampler **920**. In the graph, the vertical axis schematically represents a flow rate, and the horizontal axis represents a time.

(78) As indicated in the graph of supplement **N6**, at time **T0**, driving of first pump **911** and second pump **912** is started. Thus, supply of the first solvent and the second solvent to the analysis flow path is started.

(79) Subsequently, at time **T1** to time **T2**, the sample is introduced from autosampler **920** into mixer **913X**. The sample is thus diluted with the first solvent in mixer **913X**, and then introduced into column **931**.

(80) When the introduction of the sample is complete at time **T2**, the second solvent is introduced into mixer **913X**. Thus, the compound liquid of the first solvent and the second solvent is supplied to the analysis flow path as the mobile phase. Though the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter is constant in FIG. **7**, in actuality, the flow rate ratio between the first solvent and the second solvent at time **T2** and thereafter may change based on settings (settings according to the gradient elution method) adopted in analysis of the sample.

(81) [Comparison with Configurations According to Existing Techniques]

(82) The configuration of the embodiment of the present disclosure has been described mainly with reference to FIGS. **2** and **3**. Also, the configurations according to the existing techniques have been described with reference to FIGS. **4** to **7**. The configuration of the embodiment of the present disclosure will now be described by comparison with the configurations according to the existing techniques.

(83) FIG. **8** shows evaluations of the configuration of the present embodiment and each of the configurations according to the existing techniques in terms of five criteria. The five criteria are as follows. (1) Change of dilution factor (2) Introduction cost (3) Change of dilution mixer (4) Change of gradient mixer (5) Performance of standard injection

(84) In FIG. 8, results are indicated by ranks A to C. Rank A represents that a configuration is suitable for an analysis in terms of a criterion. Rank B represents that it cannot be said that a configuration is suitable for an analysis in terms of a criterion. Rank C represents that a configuration is not suitable for an analysis in terms of a criterion. The contents and results of the criteria will be described below.

(85) (1) Change of Dilution Factor

(86) The criterion “change of dilution factor” means whether a dilution factor of a sample can be changed in the dilution mode.

(87) In the bypass scheme (FIG. 4), the dilution factor is based on a ratio of diameter between the bypass line (pipe 814) that introduces the first solvent and the second solvent into second mixer 913D and pipe 815 that introduces the sample into second mixer 913D. In the bypass scheme (FIG. 4), the ratio of diameter between pipe 814 and pipe 815 is fixed. In the bypass scheme, accordingly, the dilution factor of the sample is fixed and cannot be changed. In the bypass scheme, thus, rank C is provided in terms of the criterion “change of dilution factor”.

(88) In both of the diluting solution introduction scheme (FIG. 5) and the sample solution introduction scheme (FIG. 6), the dilution factor of the sample can be changed by adjusting the flow rate of third pump 990. Also, in the high-pressure gradient dilution scheme (FIG. 7), the dilution factor of the sample can be changed by adjusting the flow rate of first pump 911. In the diluting solution introduction scheme, the sample solution introduction scheme, and the high-pressure gradient dilution scheme, thus, rank A is provided in terms of the criterion “change of dilution factor”.

(89) In the present embodiment, the dilution mode is shown in FIG. 3. In the configuration shown in FIG. 3, LC system 100 can change the dilution factor of the sample by adjusting the flow rate of first pump 11. Accordingly, rank A is provided to the configuration of the present embodiment in terms of the criterion “change of dilution factor”.

(90) (2) Introduction Cost

(91) The second criterion “Introduction cost” means that an increase in cost can be suppressed in implementation of the LC system.

(92) In comparison with the configurations of the bypass scheme (FIG. 4), the high-pressure gradient dilution scheme (FIG. 7), and the present embodiment (FIGS. 2 and 3), the diluting solution introduction scheme (FIG. 5) and the sample solution introduction scheme (FIG. 6) need the third pump (third pump 990).

(93) Thus, in terms of the criterion “introduction cost”, rank A is provided to the configurations of the bypass scheme (FIG. 4), the high-pressure gradient dilution scheme (FIG. 7), and the present embodiment. Contrastingly, rank C is provided to the diluting solution introduction scheme (FIG. 5) and the sample solution introduction scheme (FIG. 6).

(94) The configuration of the present embodiment needs a component for switching analysis flow path 50 between the standard position (FIG. 2) and the dilution position (FIG. 3). However, such a component can be implemented by a flow path switching valve as described with reference to FIGS. 9 and 10. The flow path switching valve needs no expensive drive unit such as a pump. Even if the component for switching analysis flow path 50 between the standard position (FIG. 2) and the dilution position (FIG. 3) is required, thus, rank A is provided to the configuration of the present embodiment in terms of the criterion “introduction cost”.

(95) (3) Change of Dilution Mixer

(96) The criterion “change of dilution mixer” means that the mixer used in dilution of the sample in the dilution mode can be changed.

(97) In all of the configurations of FIGS. 4 to 6, second mixer 913D is detachable from the analysis flow path, and is accordingly replaceable. In the configuration of FIG. 7, mixer 913X is detachable from the analysis flow path, and is accordingly replaceable. In all of the configurations of FIGS. 4 to 7, thus, the mixer used in dilution of the sample in the dilution mode can be changed.

(98) In the configurations of FIGS. 4 to 7, however when each configuration is used in the standard mode, the sample always flows through the mixer used in dilution of the sample in the dilution mode. In the configurations of FIGS. 4 to 7, thus, change (replacement) of the mixer affects the analysis performance in the standard mode.

(99) Contrastingly, such an effect is not caused in the configuration of the present embodiment. Specifically, in the configuration of the present embodiment, both of first mixer 13G and second mixer 13D are detachable from analysis flow path 50, and are accordingly replaceable. Moreover, in the standard mode, the sample introduced from autosampler 20 is not required to flow through the mixer (second mixer 13D) used in dilution of the sample in the dilution mode, as described with reference to FIG. 2. In the configuration of the present embodiment, thus, change (replacement) of second mixer 13D does not affect analysis performance in the standard mode.

(100) As described above, in terms of the criterion “change of dilution mixer”, rank B is provided to the configurations of FIGS. 4 to 7, whereas rank A is provided to the configuration of the present embodiment.

(101) (4) Change of Gradient Mixer

(102) The criterion “change of gradient mixer” means that the mixer used to mix the first solvent and the second solvent according to the gradient elution method can be changed in the standard mode.

(103) In the configurations of FIGS. 4 to 7 and the configuration of the present embodiment, the mixer is detachable from the analysis flow path. In the configurations of FIGS. 4 to 7 and the configuration of the present embodiment, thus, the mixer used to mix the first solvent and the second solvent according to the gradient elution method can be changed in the standard mode.

(104) Note that mixer 913X in the configuration of FIG. 7 (high-pressure gradient dilution scheme) is used to mix the first solvent and the second solvent and to dilute the sample. In the configuration of FIG. 7, thus, the mixer used to mix the first solvent and the second solvent cannot be changed independently of the mixer for diluting the sample.

(105) In terms of the criterion “change of gradient mixer”, thus, rank A is provided to the configurations of FIGS. 4 to 6 and the configuration of the present embodiment, whereas rank B is provided to the configuration of FIG. 7.

(106) (5) Performance of Standard Injection

(107) The criterion “performance of standard injection” means the degree of analysis performance in the standard mode.

(108) The bypass scheme (FIG. 4) is not suitable for an analysis in the standard mode. The reason for this is a behavior as follows. As long as the bypass line (pipe 814) is connected to second mixer 913D, when the sample from autosampler 20 is introduced into second mixer 913D, a part of the sample is fed through pipe 814 to pipe 813 and then flows through the sample at pipe 815 again and returns to pipe 814. In terms of the criterion “performance of standard injection”, thus, rank C is provided to the configuration of FIG. 4.

(109) It cannot be said that the diluting solution introduction scheme (FIG. 5) is suitable for the analysis in the standard mode. More specifically, it can also be said that in the diluting solution introduction scheme (FIG. 5), an analysis in the standard mode can be performed by stopping third pump 990. However, the sample flows through second mixer 913D before reaching column 931. As the sample flows through second mixer 913D, thus, the sample may be negatively affected. In terms of the criterion “performance of standard injection”, thus, rank B is provided to the configuration of FIG. 5.

(110) The sample solution introduction scheme (FIG. 6) is not suitable for the analysis in the standard mode. The reason for this is as follows. When third pump 990 is driven to feed the sample from autosampler 920 to column 931, the sample and the diluting solvent supplied from third pump 990 are mixed in second mixer 913D, and accordingly, the sample is diluted with the diluting solvent. In terms of the criterion “performance of standard injection”, thus, rank C is provided to

the configuration of FIG. 6.

(111) The high-pressure gradient dilution scheme (FIG. 7) is not suitable for the analysis in the standard mode. The reason for this is as follows. When second pump **912** is driven to feed the sample from autosampler **920** to column **931**, the sample and the second solvent supplied from second pump **912** are mixed in second mixer **913D**, and accordingly, the sample is diluted with the second solvent. In terms of the criterion “performance of standard injection”, thus, rank C is provided to the configuration of FIG. 7.

(112) Contrastingly, in the configuration of the present embodiment, the sample from autosampler **20** reaches column **31** without passing through the mixer in the analysis in the standard mode, as described with reference to FIG. 2. In terms of the criterion “performance of standard injection”, thus, rank A is provided to the configuration of the present embodiment.

(113) In terms of at least one of the five criteria described in (1) to (5) above, evaluations other than A are provided to the configurations of FIGS. 4 to 7, as described with reference to FIG. 8.

Contrastingly, an evaluation of rank A is provided to the configuration of the present embodiment in terms of all of the five criterion described in (1) to (5) above.

(114) In particular, evaluations other than rank A are provided to all of the configurations of FIGS. 4 to 7 in terms of the criterion (criterion “performance of standard injection”) described in (5), whereas an evaluation of rank A is provided to the configuration of the present embodiment.

(115) [Specific Configuration Example of Liquid Feeding Unit]

(116) FIG. 9 shows a specific configuration example of liquid feeding unit **13**. In the example of FIG. 9, liquid feeding unit **13** includes first mixer **13G**, second mixer **13D**, and a flow path switching valve **600**. Flow path switching valve **600** is implemented by, for example, a flow path switching valve manufactured by Shimadzu Corporation

(https://www.an.shimadzu.co.jp/hplc/prominence/modules/7_flow-changeover-valve.htm). Flow path switching valve **600** has six ports **601** to **606**. Each of ports **601** to **606** can be coupled to its adjacent one port. Port **606** is a port that stops a flow. Flow path switching valve **600** switches a flow path in analysis flow path **50** by changing a set of connected ports among the plurality of ports. Consequently, the flow path in analysis flow path **50** can be switched while saving space.

(117) First mixer **13G** has three openings. The first opening is connected to first pump **11** by pipe **131**. The second opening is connected to port **602** by pipe **1302**. The third opening is connected to port **605** by a pipe **1305**.

(118) Second mixer **13D** has three openings. The first opening is connected to second pump **12** by pipe **132**. The second opening is connected to port **601** by a pipe **1301**. The third opening is connected to autosampler **20** by pipe **134**.

(119) FIG. 10 shows a connection pattern example of ports **601** to **606** at a standard position of flow path switching valve **600**. As shown in FIG. 10, at the standard position, port **601** is connected to port **602**, port **603** is connected to port **604**, and port **605** is connected to port **606**.

(120) In FIG. 2, the first solvent flows through pipe **133**. Pipe **133** of FIG. 2 is implemented by pipe **1302**, port **602**, port **601**, and pipe **1301** of FIG. 10. In FIG. 2, the sample flows through pipe **135**. Pipe **135** of FIG. 2 is implemented by pipe **1304**, port **604**, port **603**, and a pipe **1303** of FIG. 10.

(121) The first solvent flows through pipe **131**, the second solvent flows through pipe **132**, and the compound liquid of the first solvent and the second solvent flows through pipe **134**.

(122) FIG. 11 shows a connection pattern example of ports **601** to **606** at the dilution position of flow path switching valve **600**.

(123) As shown in FIG. 11, at the dilution position, port **601** is connected to port **606**, port **603** is connected to port **602**, and port **605** is connected to port **604**.

(124) In FIG. 3, the sample flows through pipe **135**. Pipe **135** of FIG. 3 is implemented by pipe **1304**, port **604**, port **605**, and pipe **1305** of FIG. 11.

(125) In FIG. 3, the sample (sample+first solvent) diluted with the first solvent flows through pipe

136. Pipe **136** of FIG. **3** is implemented by pipe **1302**, port **602**, port **603**, and pipe **1303** of FIG. **11**. The first solvent flows through pipe **131**, and the second solvent flows through pipe **132** and pipe **134**.

(126) [Process Flow]

(127) FIG. **12** is a flowchart of a process performed by LC system **100** to analyze a sample. In one implementation, the process of FIG. **12** is implemented as processor **61** of controller **60** executes a prescribed program.

(128) At step **S10**, LC system **100** reads settings of an analysis mode. In one implementation, the analysis mode is registered with storage device **62**, for example, as the user operates an input device. The input device registers the analysis mode with storage device **62** via interface **63**.

(129) At step **S12**, LC system **100** determines whether the read analysis mode is the standard mode or the dilution mode. LC system **100** moves control to step **S14** when determining that the analysis mode is the standard mode and moves control to step **S16** when determining that the analysis mode is the dilution mode.

(130) At step **S14**, LC system **100** controls the state of liquid feeding unit **13** to the standard position (FIG. **2**). In one implementation, at step **S14**, controller **60** controls the connection state of ports **601** to **606** of flow path switching valve **600** to the state shown in FIG. **10**.

(131) At step **S16**, LC system **100** controls the state of liquid feeding unit **13** to the dilution position (FIG. **3**). In one implementation, at step **S16**, controller **60** controls the connection state of ports **601** to **606** of flow path switching valve **600** to the state shown in FIG. **11**.

(132) At step **S18**, LC system **100** starts flowing the mobile phase into analysis flow path **50** by starting driving of first pump **11** and second pump **12**. The control of step **S18** corresponds to a period from time **T0** to time **T1** shown in FIG. **2** and the like.

(133) At step **S20**, LC system **100** causes autosampler **20** to inject the sample into analysis flow path **50**. The control of step **S20** corresponds to a period from time **T1** to time **T2** in FIG. **2** and the like.

(134) At step **S22**, LC system **100** analyzes the sample. The control of step **S22** corresponds to a period at time **T2** and thereafter in FIG. **2** and the like. In one implementation, an analysis is performed as controller **60** obtains a detection signal based on the components of the sample separated in column **31** from detector **40** and processes the obtained detection signal. LC system **100** then ends the process of FIG. **12**.

(135) [Aspects]

(136) A person skilled in the art understands that the illustrative embodiments described above are specific examples of the following aspects.

(137) (Clause 1) A liquid feeding unit according to an aspect may be a liquid feeding unit that feeds a first solution and a second solution as a mobile phase to a column of a liquid chromatography analysis system. The first solution may be supplied from the first pump. The second solution may be supplied from the second pump. The liquid chromatography analysis system may include an injection part into which a sample is injected. The liquid feeding unit may include a first mixer, a second mixer, and a switch device that switches a flow path from the first pump and the second pump to the column between a first flow path and a second flow path. The first mixer and the second mixer may be included in each of the first flow path and the second flow path. In the first flow path, the first mixer may be located upstream of the injection part, and the second mixer may be located downstream of the injection part. In the second flow path, the first mixer and the second mixer may be located upstream of the injection part. The switch device may form the first flow path as the flow path in a first mode in which the sample is diluted before introduction into the column, and may form the second flow path as the flow path in a second mode in which the sample is not diluted before introduction into the column.

(138) According to the liquid feeding unit of the first clause, in the mode in which the sample is diluted before introduction into the column, the sample injected from the injection part can be

diluted by the second mixer. On the other hand, in the mode in which the sample is not diluted before introduction into the column, the sample can reach the column without flowing through both of the first mixer and the second mixer. Accordingly, the liquid chromatography analysis system is suitable not only for an analysis in the mode in which the sample is diluted before introduction into the column, but also for an analysis in the mode in which the sample is not diluted before introduction into the column.

(139) (Clause 2) The liquid feeding unit according to the first clause may further include a controller that controls a state of the switch device. The controller may control the state of the switch device to form the first flow path in the first mode, and may control the state of the switch device to form the second flow path in the second mode.

(140) According to the liquid feeding unit of the second clause, the state of the switch device can be controlled by the controller, thus reducing a burden for switching the state of the switch device by an operator of the liquid chromatography analysis system.

(141) (Clause 3) In the liquid feeding unit according to the first or second clause, at least one of the first mixer and the second mixer may be detachable from the flow path.

(142) According to the liquid feeding unit of the third clause, at least one of the first mixer and the second mixer can be changed.

(143) (Clause 4) In the liquid feeding unit according to the third clause, the first mixer and the second mixer may have different mixing capacities.

(144) According to the liquid feeding unit of the fourth clause, the capability of mixing solutions can be made different between the first mixer and the second mixer in accordance with a desire of the user of the liquid chromatography analysis system.

(145) (Clause 5) In the liquid feeding unit according to any one of the first to fourth clauses, the switch device may include a flow path switching valve.

(146) According to the liquid feeding unit of the fifth clause, the switch device can be implemented while saving space.

(147) (Clause 6) A liquid chromatography analysis system according to an aspect may be a liquid chromatography analysis system that uses a first solution and a second solution as a mobile phase. The liquid feeding unit may include a column, a first pump that supplies the first solution, a second pump that supplies the second solution, a sampler that injects a sample toward the column, a first mixer, a second mixer, and a switch device that switches a flow path from the first pump and the second pump to the column between a first flow path and a second flow path. The first mixer and the second mixer may be included in each of the first flow path and the second flow path. In the first flow path, the first mixer may be located upstream of an injection part into which the sample is injected by the sampler, and the second mixer may be located downstream of the injection part. In the second flow path, the first mixer and the second mixer may be located upstream of the injection part. The switch device may form the first flow path as the flow path in a first mode in which the sample is diluted before introduction into the column, and may form the second flow path as the flow path in a second mode in which the sample is not diluted before introduction into the column.

(148) According to the liquid chromatography analysis system of the sixth clause, in the mode in which the sample is diluted before introduction into the column, the sample injected from the injection part can be diluted by the second mixer. On the other hand, in the mode in which the sample is not diluted before introduction into the column, the sample can reach the column without flowing through both of the first mixer and the second mixer. The liquid chromatography analysis system is thus suitable not only for the mode in which the sample is diluted before introduction into the column but also for the analysis in the mode in which the sample is not diluted before introduction into the column.

(149) (Clause 7) The liquid chromatography analysis system according to the sixth clause may further include a controller that controls a state of the switch device. The controller may control the state of the switch device to form the first flow path in the first mode, and may control the state of

the switch device to form the second flow path in the second mode.

(150) According to the liquid chromatography analysis system of the seventh clause, the state of the switch device is controlled by the controller. This can reduce a burden for switching the state of the switch device by the operator of the liquid chromatography analysis system.

(151) (Clause 8) In the liquid chromatography analysis system according to the sixth or seventh clause, at least one of the first mixer and the second mixer may be detachable from the flow path.

(152) According to the liquid chromatography analysis system of the eighth clause, at least one of the first mixer and the second mixer can be replaced.

(153) (Clause 9)

(154) In the liquid chromatography analysis system according to the eighth clause, the first mixer and the second mixer may have different mixing capacities.

(155) According to the liquid chromatography analysis system of the ninth clause, the capability of mixing the solutions can be made different between the first mixer and the second mixer in accordance with a desire of a user of the liquid chromatography analysis system.

(156) (Clause 10) In the liquid chromatography analysis system according to any one of the sixth to ninth clauses, the switch device may include a flow path switching valve.

(157) According to the liquid chromatography analysis system of the tenth clause, the switch device can be implemented while saving space.

(158) (Clause 11)

(159) A control method according to an aspect may be a control method for a liquid chromatography analysis system that uses a first solution and a second solution as a mobile phase, and the control method may be implemented by a computer. The control method may include: obtaining an analysis mode; and controlling a flow path from a first pump and a second pump to a column of the liquid chromatography analysis system in accordance with the analysis mode, the first pump supplying the first solution, the second pump supplying the second solution, the liquid chromatography analysis system may include an injection part into which a sample is injected, a first mixer, and a second mixer, and the controlling the flow path may include: when the analysis mode is a first mode in which the sample is diluted before introduction into the column, controlling the flow path such that the first mixer is located upstream of the injection part and the second mixer is located downstream of the injection part; and when the analysis mode is a second mode in which the sample is not diluted before introduction into the column, controlling the flow path such that the first mixer and the second mixer are located upstream of the injection part.

(160) According to the control method of the eleventh clause, in the mode in which the sample is diluted before introduction into the column, the sample injected from the injection part can be diluted by the second mixer. On the other hand, in the mode in which the sample is not diluted before introduction into the column, the sample can reach the column without flowing through both of the first mixer and the second mixer. The liquid chromatography analysis system is thus suitable not only for an analysis in the mode in which the sample is diluted before introduction into the column but also for an analysis in the mode in which the sample is not diluted before introduction into the column.

(161) Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

Claims

1. A control method for a liquid chromatography analysis system that uses a first solution and a second solution as a mobile phase, the control method being implemented by a computer, the control method comprising: obtaining an analysis mode; and controlling a flow path from a first

pump and a second pump to a column of the liquid chromatography analysis system in accordance with the analysis mode, the first pump supplying the first solution, the second pump supplying the second solution, wherein the liquid chromatography analysis system includes an injection part into which a sample is injected, a first mixer, and a second mixer, and the controlling the flow path includes when the analysis mode is a first mode in which the sample is diluted before introduction into the column, controlling the flow path such that the first mixer is located upstream of the injection part and the second mixer is located downstream of the injection part, and when the analysis mode is a second mode in which the sample is not diluted before introduction into the column, controlling the flow path such that the first mixer and the second mixer are located upstream of the injection part.
